Mission Brief: Newton’s First Law

Learning sheet for exploring motion and force

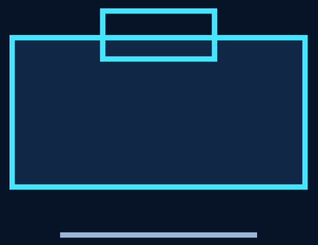
Guiding Question

- Does motion need a force to keep going?
- What really makes moving objects slow down?

Your Evidence Log

- Record what changes when surfaces get smoother.
- Use data before making conclusions.

Visual Model



A push starts motion.

Observation Station

Everyday motion can be misleading

What You See

- A suitcase slows after a push.
- A skateboard speeds up on a ramp.

What To Ask

- Which force changes the motion?
- Is the net force zero or nonzero?

Visual Model

Less resistance -> longer distance

Galileo’s Thought Experiment

From real surfaces to ideal reasoning

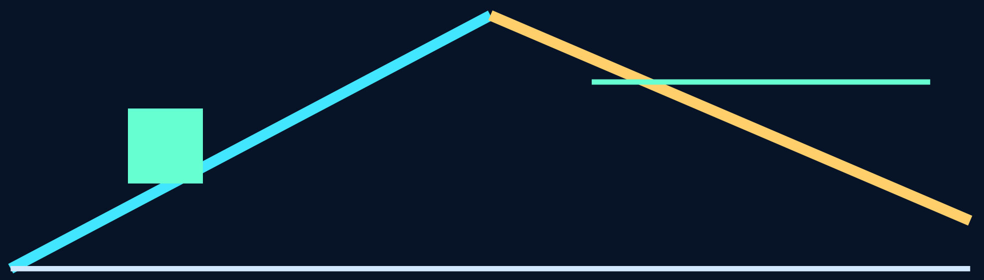
Experiment Path

- Same release height on the left ramp.
- Right ramp angle becomes smaller.

Ideal Leap

- If resistance is ignored, motion continues uniformly.
- This is reasoning beyond direct observation.

Visual Model



As the right ramp becomes flatter, the travel distance grows.

Lab Tasks

Use the simulation as a science notebook

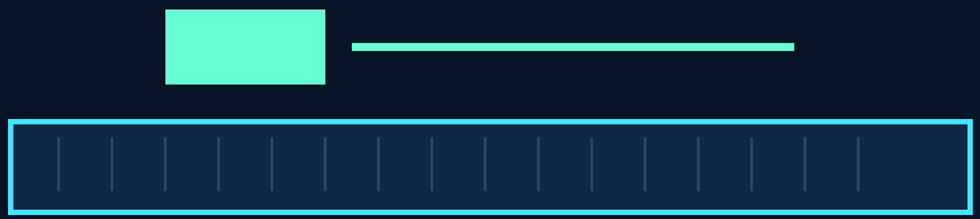
Run Three Tests

- Rough surface, smooth surface, air track.
- Compare distance and velocity graphs.

Explain With Physics

- Balanced forces keep motion unchanged.
- Nonzero net force changes motion.

Visual Model



Nearly zero net force -> nearly uniform motion